

Enhancing Customer Service Through LLM-Powered Email Drafting: An E.ON Data and Analytics Use Case

Data Analytics in Applications, TUM School of Management, WS 2025/2026

Aslihan Berber
M.Sc. Management and Technology
TUM School of Management
Munich, Germany
Matriculation: 03798114
aslihan.berber@tum.de

Mohammed Yazan Aljleilati
M.Sc. Management and Technology
TUM School of Management
Munich, Germany
Matriculation: 03728944
yazan.aljleilati@tum.de

Abstract—E.ON customer service agents handle high volumes of written inquiries, where speed, consistency, and factual correctness are critical. This project proposes a lightweight, human-in-the-loop email drafting system that combines topic modeling, short rule-based summarization, and a compact open-source language model for tone-controlled response generation. Given that the available dataset contains only incoming customer emails and no paired agent replies, we construct an instruction-tuning dataset via controlled synthetic response generation using multiple external LLMs, producing both formal and friendly targets for each email. We then fine-tune TinyLlama-1.1B with parameter-efficient QLoRA under Google Colab constraints and enforce explicit prompt-level compliance rules together with deterministic decoding to reduce unreliable outputs. In a small held-out evaluation, the system generates coherent drafts with stable training behavior, moderate lexical and semantic variation, and no detected introduction of new factual entities under a rule-based hallucination check. Finally, we deploy the workflow as a Streamlit prototype that presents topic labels, short summaries, and dual-tone drafts to support agent review and finalization.

Index Terms—Customer service, email drafting, large language models, TinyLlama, QLoRA, instruction fine-tuning, BERTopic, human-in-the-loop, safety constraints.

I. INTRODUCTION

Customer service operations rely heavily on written communication, particularly in industries with large and diverse customer bases such as the energy sector. At E.ON, customer inquiry emails cover recurring topics including billing and payments, meter readings, contract changes, online portal access, and supply-related issues. In businesses, professionals spend a substantial portion of their working time handling email communication, making it a significant source of operational overhead (Thiergart et al., 2021). The practical challenge for organizations like E.ON is therefore not to fully automate customer communication, but to support customer service agents by accelerating the drafting process, improving consistency, and reducing repetitive manual effort while maintaining human oversight.

Recent advances in Natural Language Processing (NLP), particularly through Large Language Models (LLMs), have fundamentally reshaped the automation of text-based tasks. LLMs differ from earlier NLP approaches primarily due to their scale and training on large, diverse datasets, which enables emergent capabilities such as instruction following, contextual reasoning, and fluent text generation (Cai et al., 2025). These capabilities have rapidly translated into real-world applications. Commercial systems such as GitHub Copilot, conversational search engines, and meeting summarization tools demonstrate how LLMs can act as flexible language-based interfaces for complex knowledge work (Ünlü, 2023). Prior studies show that LLM-assisted email drafting is technically and economically viable and can meaningfully support customer service workflows when used as a drafting aid rather than a fully autonomous system (Thiergart et al., 2021; Wulf and Meierhofer, 2024).

In parallel, current literature highlights the importance of structuring unstructured customer communication to support downstream automation. Topic detection and clustering techniques based on contextual embeddings have been shown to outperform traditional statistical models when applied to short, noisy texts such as emails (Dahl et al., 2025; Janssens et al., 2025). In particular, BERTopic (Grootendorst, 2022) and related BERT-based approaches enable the extraction of interpretable, operationally meaningful topics, which can support routing, prioritization, and response generation in customer service settings. These findings motivate the integration of topic modeling as an explicit component of our project and proposed email drafting tool.

The implementation of the project is carried out entirely in Python and all experiments are executed in a Google Colab environment under constrained GPU memory and limited runtime, which shaped both the modeling choices and the overall system design. In particular, these constraints ruled out training or deploying very large proprietary foundation models

within the scope of this project and encouraged a compact, reproducible pipeline that can run on commonly accessible hardware.

Rather than treating these constraints purely as limitations, we frame them as realistic enterprise conditions and an explicit part of the research setup. The project therefore investigates whether a small open-source language model, combined with parameter-efficient fine-tuning and conservative decoding, can deliver useful drafting support for customer service emails without requiring large-scale computation or extensive labeled data. This aligns with our moderate size of available dataset and supports the broader objective of assessing lightweight, cost-efficient LLM deployments for applied business workflows.

We utilize *TinyLlama/TinyLlama-1.1B-Chat-v1.0* as the base model and adapt it to the E.ON email domain using parameter-efficient fine-tuning. However, this project also addresses a central constraint that is common in enterprise settings: the dataset contains only incoming customer emails and does not include paired agent replies. Since supervised instruction fine-tuning typically relies on instruction-response pairs, we construct such pairs through controlled synthetic response generation. Using multiple external LLMs, each customer email is converted into a standardized instruction format with two target outputs (formal and friendly), forming an instruction-tuning dataset that preserves domain relevance while explicitly modeling tone variation.

The resulting system combines topic detection, lightweight rule-based summarization, and LLM-based response drafting into a single end-to-end workflow. Incoming customer emails are first assigned to one of the predefined BERTopic categories and are summarized by removing the greetings and extracting the most informative sentences. The fine-tuned TinyLlama model then generates two tone-controlled draft replies. To reduce the risk of unreliable outputs, response generation is constrained through explicit prompt-level compliance rules and deterministic decoding, ensuring that replies remain grounded in the customer-provided content and avoid invented factual details. Consistent with recommendations from both academic and industry literature, the system is designed as a human-in-the-loop drafting aid, where agents review and finalize all outgoing communication (BCG, 2023; Costa and Ghosh, 2025).

The contributions of this work are threefold. First, we present an end-to-end pipeline that transforms unpaired real-world customer emails into an instruction-tuning dataset via controlled synthetic response generation and integrates topic modeling and summarization to add structure for downstream drafting. Second, we demonstrate that a compact open-source language model can be adapted to customer service email drafting under realistic computational constraints using parameter-efficient fine-tuning. Third, we implement and evaluate a prototype Streamlit web application that operationalizes the workflow by combining topic labeling, short summarization, and dual-tone response drafting as decision support for customer service agents all inside the Google

Colab environment.

The remainder of this paper is structured as follows; Section II introduces the business context and motivation of the E.ON use case, followed by data understanding and topic modeling. Sections thereafter describe synthetic data generation, model implementation and fine-tuning, evaluation, and the deployed web application, before concluding with a discussion of limitations and future work.

II. BUSINESS UNDERSTANDING

A. E.ON's Position

E.ON is an investor-owned energy company that employs around 76,600 full-time equivalent employees, led by the corporate functions department in Essen. Since the beginning of 2024, the Group's core business has been divided into three business divisions: Energy Networks, Energy Infrastructure Solutions and Energy Retail.

It offers sustainable solutions for increased energy efficiency, autonomy, and e-Mobility while supplying conventional and green gas and electricity to clients throughout Europe. It is separated into Germany, the UK, the Netherlands, and other regions, including Sweden, Italy, the Czech Republic, Hungary, Croatia, Romania, and Poland, and caters to a variety of customer categories, such as residential, SMEs, major commercial and industrial clients, and sales partners. E.ON wants to actively support Europe's energy transition while also improving consumer satisfaction (E.ON SE, 2024).

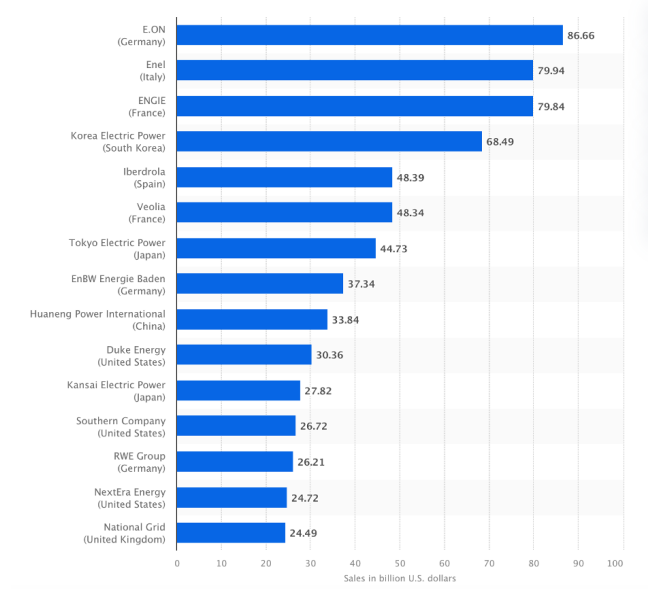


Fig. 1. Sales revenue of largest electric utility companies worldwide 2025 (Forbes, 2025)

A global sales comparison also puts E.ON's relevance in the market into perspective. As shown in Figure 1, E.ON recorded the highest sales among major international electric utility companies, reaching 86.66 billion U.S. dollars. This positions the company ahead of other large competitors such as Enel, which reported 79.94 billion U.S. dollars, followed by ENGIE,

which reported 79.84 billion U.S. dollars. The strong revenue base shows E.ON's international strategic importance in the energy field and provides scale for its customer operations.

B. Context & Use Case Description

As one of the top five biggest energy suppliers, E.ON receives hundreds of customer emails per day covering billing, service interruptions, meter queries, contract changes, and technical support. The operational need is not to fully automate every reply immediately, but to provide a robust framework that generates reusable answer components that customer agents or automated systems can assemble into final replies. This will help customer agents generate a suitable response to each customer email, saving time by partly automating the process, instead of writing a response to each email individually.

C. Related Business Use Cases in Customer Email Drafting

Industries with heavy customer contact have begun leveraging NLP and LLM-based systems to streamline email communications. Early approaches to automated email replies date back several years. Google's "Smart Reply" system was deployed in Gmail to suggest short responses, and by 2016 it was already assisting with about 10% of all mobile email replies by generating plausible one-tap answers (Kannan et al., 2016). This demonstrated the feasibility of AI-driven email suggestions at scale. Recent advancements in large language models have further expanded these capabilities. A 2021 study showed that using an LLM (GPT-3) to comprehend customer emails and draft responses is technically and economically viable, indicating that modern models can understand incoming messages and formulate appropriate replies in business settings (Thiergart et al., 2021). Similarly, Wulf and Meierhofer (2023) demonstrated prototypes where GPT-4-class models summarized customer inquiries and answered questions in a technical support domain, highlighting both the potential efficiency gains and the new challenges, such as ensuring factual accuracy and data privacy, of integrating LLMs into customer service workflows. Overall, the current literature confirms that AI-powered email drafting, especially with cutting-edge LLMs, is now practical for handling high volumes of customer communication.

A closer look can be taken into the related academic contribution of the work by Thiergart et al. (2021), who provide one of the earliest formal analyses of using a large language model (GPT-3) for understanding incoming emails and generating draft responses. Their study demonstrates that GPT-3 can reliably execute tasks such as intent recognition, summarization, and natural-language reply drafting without fine-tuning, making it a versatile technology for organizations dealing with large volumes of customer emails. They further emphasize that email handling constitutes a significant productivity drain, with professionals spending nearly 28% of their workweek on email management. Importantly, they identify limitations such as hallucinations, restricted access to business-internal knowledge, and the need for high-quality

prompts. To address these constraints, the authors propose strategies including rule-based guardrails, context injection, and mandatory human review.

Real-world case studies across industries have also begun to validate these findings. In the energy sector, Octopus Energy, a UK-based utility, reported significant success after integrating generative AI into its customer contact management. The AI system helps agents draft rich, detailed email responses at a speed and thoroughness previously unattainable. According to a Boston Consulting Group analysis, AI-drafted emails at Octopus achieved 18% higher customer satisfaction scores compared to human-written replies, and the system now automatically handles roughly one-third of all customer inquiry emails, freeing human agents to focus on complex or high-value cases (BCG, 2023). This result illustrates how an LLM-based assistant can maintain or improve quality while substantially reducing agent workload.

Additional business deployments from other industries further reinforce these observations. Klarna (Klarna Group plc) has reported significant efficiency gains after integrating LLM-based automation into customer communication workflows, enabling faster resolution of repetitive inquiries and reducing operational overhead (Costa & Ghosh, 2025). Similarly, SAP (SAP SE) has leveraged generative AI to support customer service agents through automated summarization, answer drafting, and interactive guidance, demonstrating how LLMs can act as "co-pilots" that augment, rather than replace, human decision-making (Costa & Ghosh, 2025). These examples highlight a broader trend across sectors, where companies use generative AI to accelerate routine communication tasks while maintaining human oversight for accuracy and customer trust.

The need for human-AI collaboration in customer service is further reinforced by Costa and Ghosh (2025), who examine how organizations integrate generative AI into their support operations. Their findings show that while generative AI significantly reduces cognitive load and improves handling time, customers still strongly prefer human interaction for complex or sensitive cases. Thus, they argue for a "trust-but-verify" model in which AI accelerates routine tasks and information retrieval, but human agents remain responsible for oversight, judgment, and final communication quality (Costa & Ghosh, 2025). This mirrors guidelines from industry analyses, such as those from BCG, where nearly all real-world AI support systems retain a human-in-the-loop to validate accuracy and handle exception scenarios (BCG, 2023). Together, these studies highlight that the optimal implementation of generative AI in contact centers is a semi-automated workflow in which AI drafts or synthesizes information and human agents review, edit, or approve the final output.

Ensuring factual accuracy is also a crucial challenge, particularly in regulated industries like energy where incorrect information can have significant consequences. In this regard, recent research on Retrieval-Augmented Generation (RAG) provides valuable insights for enterprise-grade email drafting systems. Li et al. (2025) show that RAG architectures can substantially improve correctness by grounding generative

outputs in retrieved, validated domain documents. Their study identifies several best practices relevant to customer email drafting, such as query expansion (allowing the LLM to refine its own search query), sentence-level “focus mode” retrieval for sharper factual grounding, and careful tuning of chunk size and retrieval stride to balance precision with contextual completeness.

In summary, both academic literature and industry case studies consistently demonstrate that LLM-based email drafting systems can enhance productivity, reduce workload, and improve the quality and consistency of customer communications. However, the evidence also clearly shows that such systems are most effective when implemented with proper oversight mechanisms, human review, and domain-grounding techniques such as RAG. This combination of AI-generated drafts plus human verification represents the most promising and responsible approach for organizations like E.ON seeking to modernize their customer contact management.

III. DATA UNDERSTANDING

A. Utilized Dataset

The dataset contains 161 anonymized customer emails provided by E.ON. These emails represent real customer requests in the form of emails written in German and cover a range of topics, including billing, contract changes and meter readings. Out of originally 181 emails, only 161 were provided since a subset was removed due to sensitive information that could not be disclosed. These remaining 161 emails form a dataset suitable for natural language processing models. The dataset contains only customer-side communications and does not include responses from customer agents.

B. Data Structure

Each email in the dataset is stored as an individual plain-text (txt) file containing one complete customer inquiry, with no embedded formatting. This allows for flexible data preparation and analysis. Personal information such as names, addresses, account details were removed or replaced beforehand to remain in compliance with data protection requirements. The files are organized in a flat directory structure, with file names following a simple incremental integer scheme (e.g., 0.txt, 1.txt, 2.txt) with some numbers missing due to the removed subset from the original cluster.

IV. DATA PREPARATION & TOPIC MODELING

A. Initial Data Ingestion and Minimal Preprocessing

After reviewing the dataset, all email files are read using a directory scan and for each file, the text content and its identifier are stored in a table with two fields: `email_id` and `raw_text`. A preview of the loaded dataset is shown in Table I.

email_id	raw_text (excerpt)
22	Seit Februar haben Sie nicht eine einzige ...
103	Hallo,\nAnbei meine Vertragsbestätigung ...
12	Guten Tag\nEs wurde noch ein Zählerfoto ...
41	Hiermit schicke ich die Unterlagen für den ...
102	Hallo EON,\nBitte erstatten Sie ...

TABLE I
EXCERPT OF RAW EMAILS AFTER INITIAL INGESTION

To maintain compatibility with downstream processing, all emails are then exported to a JSON Lines file, with one email per line. No aggressive cleaning, such as lowercasing, stemming, or lemmatization, is applied, as modern transformer-based embedding models perform best when provided with full, natural text containing capitalization and punctuation. This design choice follows findings from BERT-based research indicating that excessive normalization can degrade semantic information in contextual embeddings (Devlin et al., 2019; Siino et al., 2024).

However, stop-word removal is essential to reduce noise during topic extraction. A combined stop-word list is constructed using German stop-words from NLTK library (Bird et al., 2009) and additional domain-specific stop-words tailored to customer email conventions (e.g., “hallo”, “guten tag”, “freundlichen grüßen”, “eon”, “kundenservice”, etc.). This ensured that frequently occurring but semantically irrelevant greetings or sign-off phrases did not dominate topic keywords. The final stop-word set is integrated directly into the `CountVectorizer` used by BERTopic.

B. Topic Modeling and Clustering

Given the objective of identifying the main topics in German customer emails, BERTopic is selected as the primary topic modeling framework (Grootendorst, 2022). BERTopic combines transformer-based embeddings, non-linear dimensionality reduction, and density-based clustering to produce coherent topic groups, making it particularly effective for short or noisy real-world texts such as emails. Empirical evaluations from literature show that BERT-based embedding methods yield more coherent and semantically meaningful topics than traditional statistical models, particularly in low-resource and domain-specific settings (Janssens et al., 2025; Islam et al., 2025).

Embedding Model: To generate multilingual sentence embeddings suitable for German text, the model `paraphrase-multilingual-mpnet-base-v2` from `SentenceTransformers` is used. This model produces high-dimensional semantic representations enabling BERTopic to cluster emails based on meaning rather than surface-level word overlap (Reimers & Gurevych, 2019).

Vectorization and Dimensionality Reduction: A custom `CountVectorizer` is employed with:

- `stopwords` = combined German + domain-specific list,
- `n-grams` = (1, 2), capturing both single words and key phrases,
- `min_df` = 2 to eliminate noise from rare terms.

This is followed by UMAP (McInnes et al., 2018) as the dimensionality reduction algorithm (handled internally by BERTopic) and HDBSCAN for clustering (McInnes et al., 2017). These choices again, align with recommended BERTopic best practices for small to medium-sized datasets.

Clustering Results: Fitting BERTopic to the 161 emails yielded 15 distinct clusters in total, consisting of 14 substantive topics and one outlier group (Topic -1, *Sonstiges*). This is achieved according to:

- 1) algorithmic filtering based on word frequency, token validity, and stopword removal;
- 2) manual inspection to ensure interpretability and domain relevance.

The final topic labels are shown in Table II.

topic_id	topic_label	count
-1	Sonstiges	30
0	Überweisung / Rückzahlung	16
1	Stromversorgung	15
2	Bestätigung	13
3	Strommeldung	13
4	Zählerstand / Zählerplatz	12
5	Vertragspartner / Bestätigung	8
6	Stromzähler-Anfrage	8
7	Rechnungsanfrage	7
8	Vertragsnummer ändern	7
9	Online-Verträge / Portal	6
10	Daten & Vertragskonto	6
11	Portal-Zugang / E-Mail-Probleme	6
12	Stromvertrag / Grundvertrag	5
13	Vertrag allgemein	5
14	Zählerunterlagen / Fehler	4

TABLE II
FINAL TOPIC ASSIGNMENTS AND FREQUENCY ACROSS EMAILS

Each email is then assigned a topic and an associated confidence score derived from the probability distribution generated by BERTopic. A sample of these assignments is shown in Table III.

email_id	topic_id	topic_label	confidence
22	9	Online-Verträge / Portal	0.194
103	10	Daten & Vertragskonto	1.000
12	1	Stromversorgung	0.611
41	12	Stromvertrag / Grundvertrag	1.000
102	0	Überweisung / Rückzahlung	0.263

TABLE III
EXAMPLE OF TOPIC ASSIGNMENTS WITH CONFIDENCE SCORES

A visualization of the final topic distribution is provided in Figure 2, illustrating that the most common themes relate to payment queries, confirmations, meter readings, and account/contract changes. This confirms that customer emails cluster naturally into operationally meaningful categories, which can subsequently support downstream tasks such as automated response generation.

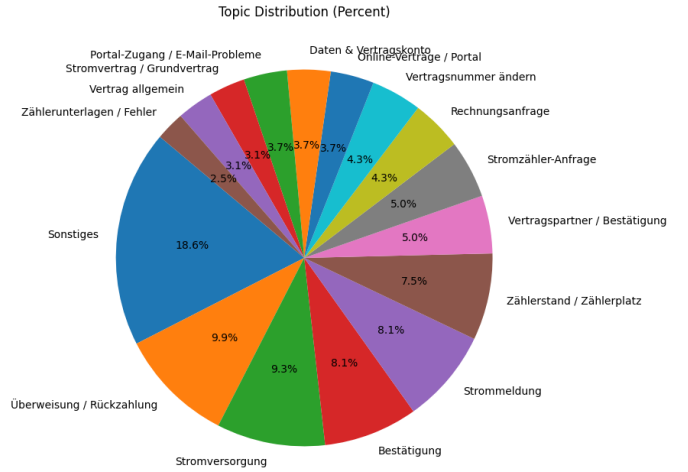


Fig. 2. Topic distribution across all emails

Overall, the BERTopic-based pipeline successfully extracts interpretable and domain-relevant clusters from a relatively small corpus of German customer emails. The combination of multilingual sentence embeddings, custom stop-word filtering, and manual refinement ensure that the resulting topics accurately reflect common customer concerns at E.ON.

V. SYNTHETIC DATA GENERATION

A. The Missing Response Problem

A major challenge in the dataset is that the customer emails do not have any corresponding customer agent responses. This creates a structural limitation as supervised fine-tuning normally requires instruction-response pairs (Ouyang et al., 2022). Without response examples, the model cannot learn the tune, structure, or problem-solving behavior expected in E.ON’s customer communication. Therefore, the dataset in its original form is insufficient for training a response generation model. Instead of applying what it looks like directly to our response generation model, synthetic data generation for the missing corresponding responses is necessary.

B. The Instruction-Tuning Data

To make the data suitable for training an LLM model, we convert all samples into a standardized instruction-response format. In the instruction, we assign the model a predefined role (“Du bist ein KI-Assistent, der Mitarbeitende im E.ON-Kundenservice unterstützt.”). This role conditioning ensures that the fine-tuned model follows E.ON’s expected communication style and avoids generating unauthorized commitments or fabricated customer information. An additional command specifies that all responses must be generated in German, as the majority of E.ON’s customers communicate exclusively in German. We also include an instruction requesting replies to be generated in a friendly or formal style. This allows customer agents to select the tone that best matches the communication style of the incoming email. The original contents of customer

emails are used as the input field. Due to data protection and privacy concerns, all names, addresses and identifiable details are replaced with anonymized placeholders. In the output field, synthetic responses are applied as the learning reference. As described previously, in our design, each email receives two versions of synthetic replies with two separate tones; one formal and one friendly, so that the final generated email can be aligned with customers' communication tone. With these elements combined, the dataset becomes suitable for instruction tuning and then fine-tuning.

C. Synthetic Response Generation

To generate the missing response component of our training data, we employ a prompt-based workflow using three different LLMs (Gemini 3 (Google, 2026), ChatGPT 5.2 (OpenAI, 2025), and DeepSeek V3 (DeepSeekAI, 2024)). Approximately each one-third section of the dataset is processed with these three LLMs, ensuring variation in model behavior and reducing model-specific response patterns.

First, a role-based prompt is initialized to assign the model the role of an E.ON customer service agent. The original customer email is then inserted as the input. The selected LLM tool is instructed to produce two versions of the reply: one formal and one friendly. The prompt explicitly requires German output and strictly prohibits hallucinations, fabricated contract details, or unrealistic procedures.

For each email, both replies are generated and manually reviewed. Outputs containing implausible details, incorrect tone, or deviations from the given information are discarded and regenerated. After validation, the replies are stored in a standardized instruction-input-output structure, resulting in two training samples per customer email.

Through this workflow, the missing response data is synthetically generated in a controlled and consistent way, producing a dataset suitable for downstream instruction tuning and model fine-tuning.

At the end of this workflow, the final full dataset ready for model training consisted of 322 emails (input and response pairs) in total.

VI. PROJECT WORKFLOW

Figure 3 summarizes the end-to-end workflow of the proposed system, from raw customer inquiry emails to assisted draft response generation. Incoming anonymized emails are first ingested and structured, followed by topic detection using BERTopic and the generation of synthetic response pairs for instruction tuning. The fine-tuned TinyLlama model is then trained and validated under explicit safety and hallucination constraints. Finally, the resulting model is deployed within a Streamlit web application, where customer service agents can generate and review structured draft responses in a human-in-the-loop setting.

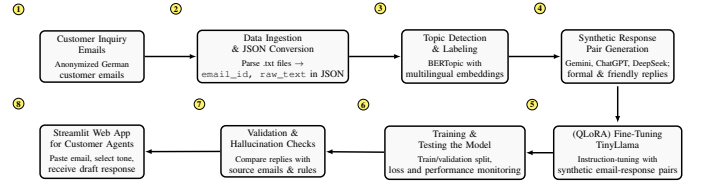


Fig. 3. End-to-end project workflow: from raw customer inquiry emails to a fine-tuned TinyLlama model and deployed Streamlit web application for assisted email response drafting.

VII. MODEL IMPLEMENTATION AND FINE TUNING

A. Base Model Selection and Computational Constraints

Fundamentally, hardware limitations and the exploratory nature of the project primarily drives our selection of the base large language model. All experiments are conducted in a Google Colab environment with restricted GPU memory and runtime, which rules out the use of very large foundation models. Rather than viewing this purely as a limitation, we treat it as a deliberate design choice to investigate how effectively a compact language model can support customer service tasks under realistic conditions. The project therefore focuses on evaluating whether a compact, open-source language model can meaningfully support customer service email drafting under realistic enterprise conditions, where computational resources and labeled data are often limited. This choice also aligns with the moderate size of the available dataset and supports the broader objective of assessing the feasibility of lightweight, cost-efficient LLM deployments for applied business use cases.

Following this, we select *TinyLlama/TinyLlama-1.1B-Chat-v1.0* as the base model (TinyLlama, 2024). With approximately 1.1 billion parameters, TinyLlama is lightweight enough to be fine-tuned on consumer-grade GPUs while still being based on a modern transformer architecture. We use the chat-aligned variant of TinyLlama, which has been pre-trained to follow natural-language instructions and generate assistant-style responses (Zhang et al., 2024). This prior alignment reduces the amount of task-specific data required and makes the model well suited for drafting structured customer service emails. By choosing a small open-source model, we aim to evaluate whether acceptable drafting quality can be achieved without relying on large proprietary models or extensive computing resources.

B. Instruction Fine-Tuning and Parameter-Efficient Optimization

To adapt the selected base model to the E.ON customer service domain, we apply supervised instruction fine-tuning using a parameter-efficient optimization strategy using the `peft` and `transformers` libraries (Mangrulkar et al., 2022; Wolf et al., 2020). Updating all model parameters is infeasible under the available hardware constraints; therefore, Low-Rank Adaptation (LoRA) is employed. LoRA introduces a limited number

of trainable parameters into selected transformer layers while keeping the original base model weights frozen, substantially reducing both memory usage and computational cost (Hu et al., 2021).

In addition, the base model is loaded in 4-bit quantized form using the `bitsandbytes` library. This quantization compresses model weights while preserving most of the representational capacity of the original parameters. The combination of LoRA with 4-bit quantization, commonly referred to as QLoRA (Dettmers et al., 2023), represents a state-of-the-art approach for fine-tuning language models under constrained hardware conditions. Applying QLoRA enables all fine-tuning steps to be executed within a single Google Colab environment.

The instruction-tuning dataset is derived from the synthetic response generation process described earlier. In total, 322 instruction-response pairs are constructed. Each original customer email is converted into two separate training examples: one requiring a formal response and one requiring a friendly response. Each sample follows a standardized instruction format that explicitly defines the model’s role as an assistant supporting E.ON customer service staff and specifies the desired communication tone. The original customer email serves as the input, and the corresponding synthetic response is used as the target output.

Dropout is applied at the LoRA adapter level to mitigate overfitting on the relatively small and synthetically generated instruction-tuning dataset. By introducing stochastic regularization during training, dropout encourages the model to learn more robust and generalizable response patterns rather than memorizing specific phrasings (Srivastava et al., 2014), which is particularly important given the repetitive instruction structure and dual-tone (formal and friendly) response setup. A low dropout rate of 0.05 is selected to balance regularization benefits with training stability under quantized fine-tuning (PyTorch Foundation).

C. Training Setup and Optimization

All training and evaluation experiments are conducted in a pay-as-you-go Google Colab environment. To obtain sufficient GPU resources for fine-tuning, we use an NVIDIA L4 GPU provided through Colab’s subscription. While more powerful GPUs are available in dedicated research or enterprise settings, the L4 represents a realistic and commonly accessible resource, reinforcing the practical constraints under which the model is developed.

The dataset is split into training and validation subsets using an **80/20 split**, resulting in **257 training samples** and **65 validation samples**. Fine-tuning is performed for a fixed number of **600 update steps** to balance learning progress with the risk of overfitting on the relatively small synthetic dataset.

A learning rate of 2×10^{-5} is used together with gradient accumulation to achieve an effective batch size of **8**. Given the limited size of the dataset and the synthetic nature of the responses, we deliberately constrain training to a fixed number of optimization steps rather than training for many

full epochs. In particular, the model is trained for **600 update steps**, corresponding to approximately 18 effective passes over the training data.

To prioritize training stability, we apply gradient clipping with a maximum norm of **0.3** (Pascanu et al., 2013), employ gradient accumulation over **8 steps**, and use a warmup ratio of **0.05** to stabilize early training updates. With a per-device batch size of 1, this resulted in an effective batch size of 8 while remaining within GPU memory constraints. The `paged_adamw_8bit` optimizer is used, which is commonly employed in QLoRA-based fine-tuning to reduce memory consumption (Dettmers et al., 2023). Mixed-precision training is disabled (`fp16=False`, `bf16=False`) to avoid numerical instability in the Colab environment.

Training progress is logged every **25** steps, and model checkpoints are saved every **300** steps, with a maximum of **2** checkpoints retained. Training stability is monitored throughout, and no numerical instabilities such as NaN or infinite values are observed in the LoRA adapter parameters.

D. Evaluation Results

After training, the model is evaluated on the held-out validation set containing **65 samples** that were not seen during training. The final training loss is **1.48**, and the validation loss is **1.30**. The validation loss being slightly lower than the training loss can be explained by the use of dropout during training, which is disabled during evaluation, resulting in noisier optimization conditions during training (Srivastava et al., 2014).

Although the compact model size and limited dataset are constraints affecting the best performance, the observed results indicate that the model successfully learned to map customer emails and structured instructions to coherent response drafts while maintaining stable training dynamics under realistic computational constraints.

TABLE IV
MODEL TRAINING CONFIGURATION AND EVALUATION METRICS

Metric	Value
Total Samples	322
Train / Validation Split	257 / 65
Training Steps	600
Learning Rate	2×10^{-5}
Effective Batch Size	8
Training Loss	1.48
Validation Loss	1.30

E. Inference Behavior and Model Output

For functional testing, this fine-tuned model is applied to and tested on 10 randomly selected customer emails that were not part of the training data. During inference, the system produces multiple structured outputs for each incoming email. First, the email is passed to the previously trained BERTopic model, which assigns it to one of the predefined topic categories identified during topic modeling. No new topics are generated at this stage; instead, the existing topic

labels are reused to ensure consistency and interpretability across the pipeline.

In addition to topic assignment, a short summary of the customer email is generated using a lightweight, rule-based approach. Common greetings, signatures, and other boilerplate expressions are removed, after which the first one to two informative sentences are extracted. This summary provides customer service agents with a concise overview of the request without relying on generative modeling.

Following topic assignment and summarization, the fine-tuned language model generates draft email responses. To ensure consistent and reproducible outputs, we use deterministic decoding rather than random sampling (Meister et al., 2020; Holtzman et al., 2020). In practice, this involves applying beam search with three beams, together with repetition control mechanisms that discourage redundant phrasing and repeated text segments. For each customer email, the system produces two versions of the reply: a formal version and a friendly version, enabling agents to select the tone that best matches the customer’s communication style.

During inference, we further constrain the response generation through hard-coded explicit instruction and rule-based prompting to ensure compliance, tone consistency, and factual correctness. Each prompt combines tone-specific style guidelines with a fixed set of safety and compliance rules. These constraints instructs the model to respond exclusively based on the information contained in the customer email, reuse numerical information exactly as written, and refrain from guessing when information was missing. If required information is uncertain, the model is instructed to ask targeted follow-up questions rather than inventing details.

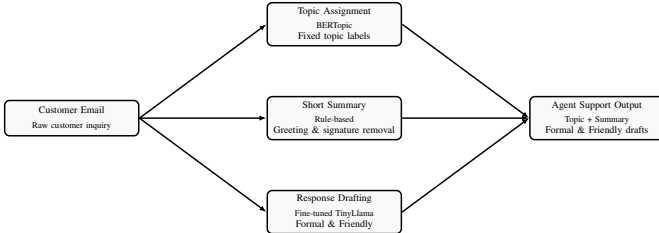


Fig. 4. Inference pipeline for customer email handling. Incoming emails are processed through topic assignment, rule-based summarization, and response drafting using the fine-tuned TinyLlama model. All components contribute to a structured agent support output combining topic labels, summaries, and tone-controlled draft responses.

VIII. EVALUATION FRAMEWORKS

A. Lexical Diversity and Semantic Variety Check

Beyond factual correctness, a practical drafting assistant must avoid producing overly repetitive or template-like responses. In customer service settings, excessive repetition can reduce perceived quality, limit adaptability to individual customer cases, and increase the manual editing burden for agents (Holtzman et al., 2020). Therefore, we evaluate whether the fine-tuned model generates responses that are both diverse in wording and semantically consistent.

To assess response diversity, we implement a two-part validation approach combining lexical entropy analysis and semantic similarity measurement. This dual strategy allows us to distinguish between superficial variation in wording and meaningful variation in content.

a) Lexical Entropy Analysis: Lexical entropy is used as a measure of surface-level linguistic diversity. For each generated response, token frequencies are computed and transformed into an entropy score. Higher entropy values indicate a broader and more balanced vocabulary distribution, while very low entropy values suggest repetitive or formulaic language (Guo et al., 2024).

The analysis is conducted on the combined set of 10 randomly selected emails with formal and friendly responses generated by the fine-tuned saved model. The observed mean lexical entropy is **3.49**, with a standard deviation of **0.15**. As illustrated in Figure 5, the entropy values form a compact but non-degenerate distribution, indicating consistent variation across responses rather than collapse into a single template.

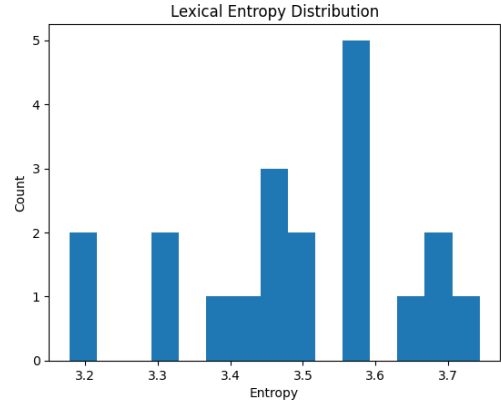


Fig. 5. Distribution of lexical entropy across generated responses.

b) Semantic Similarity Analysis: While lexical entropy captures surface variation, it does not guarantee semantic consistency (Farquhar et al., 2024). To evaluate whether responses remain aligned in meaning while still varying in phrasing, we compute pairwise semantic similarity scores using sentence embeddings generated by the `all-MiniLM-L6-v2` model. Cosine similarity is calculated for all unique response pairs, excluding self-comparisons (Reimers & Gurevych, 2019).

The resulting similarity distribution shows a mean cosine similarity of **0.63** with a standard deviation of **0.12**. As shown in Figure 6, most response pairs fall within a moderate similarity range, indicating that the model preserves core semantic intent while allowing for linguistic variation in structure and phrasing. This level of semantic similarity is expected given several characteristics of the experimental setup. First, the evaluation is conducted on a small set of ten customer emails, which inherently limits topical and contextual diversity. Many customer inquiries share similar structural patterns and intent (e.g., document submission, confirmation requests, or status updates), leading to naturally higher semantic overlap across

generated responses. Second, the model operates without retrieval-augmented generation and is explicitly constrained through hard-coded instruction prompts and safety rules that enforce factual grounding and discourage speculative content. These constraints intentionally narrow the semantic output space to ensure reliability and compliance in a regulated customer service setting.

As a result, higher semantic similarity should not be interpreted as response collapse or lack of expressiveness, but rather as an indication that the model consistently adheres to the intended meaning of the customer request while varying surface-level phrasing. In this context, moderate-to-high cosine similarity reflects controlled semantic alignment rather than redundancy, which is a desirable property for assisted email drafting tasks.

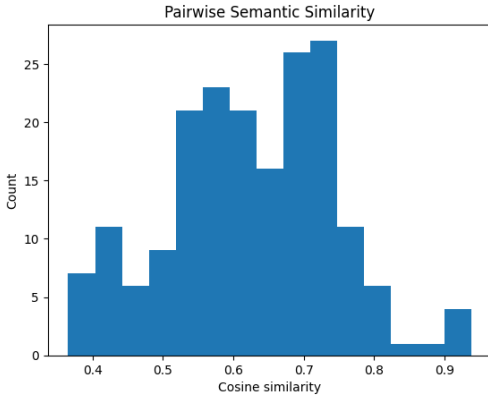


Fig. 6. Pairwise semantic similarity distribution for generated responses.

B. Hallucination Check

In customer service settings within regulated industries such as the energy sector, factual correctness is a critical requirement. Incorrectly generated numbers, dates, contact details, or URLs may result in billing errors, regulatory violations, or legal exposure. For this reason, hallucination risk analysis constitutes a core component of validating large language models intended for customer-facing drafting tasks.

To assess factual grounding, we implement a transparent, rule-based hallucination detection framework that compares each generated response directly against the corresponding customer email. Rather than relying on semantic similarity or subjective human judgment alone, this evaluation explicitly targets the introduction of new fact-like content that was not present in the original customer message. The analysis focuses on five categories of sensitive factual entities that are particularly critical in customer communication: numerical values, dates, email addresses, URLs, and phone numbers. Candidate entities are extracted from both the customer email and the generated response using regular expression patterns. Numerical expressions are normalized to account for formatting differences such as spacing or separators, ensuring that semantically identical values (e.g., “232 671” vs. “232671”) are correctly matched.

After extraction, per-response hallucination risk is detected by computing the set difference between factual entities extracted from the generated response and those present in the original customer email. If this difference is non-empty (i.e. if it introduces any new factual entity that does not appear in the original customer email), the response is flagged as a hallucination. The hallucination analysis is applied separately to the formal and friendly responses generated by the fine-tuned model. Again on the same final evaluation set of ten previously unseen customer emails, the observed hallucination rate is **0%** for both formal and friendly responses. No generated reply introduced new numbers, dates, contact details, or URLs that were not explicitly present in the corresponding customer input.

These results indicate that the model reliably conditions its outputs on customer-provided information rather than inventing new facts. However, this low hallucination rate must be interpreted in context. First, our response generation is strongly constrained by explicit, hard-coded instruction and compliance rules that prohibit factual invention and enforce strict reuse of input information. Second, the evaluation is conducted on a limited sample of ten emails, which restricts the statistical generalizability of the observed rate. The low hallucination therefore reflects the combined effect of conservative decoding strategies, hard-coded safety constraints, and post-generation validation rather than unconstrained generative behavior. This framework is designed for controlled internal evaluation and provides a transparent and practical post-training validation mechanism, with all generated outputs intended to still be subject to mandatory human review.

IX. WEB APPLICATION & DEPLOYMENT

A. System Architecture and Deployment

A prototype web application is implemented using Streamlit to provide an accessible interface for customer service agents. The application is designed as a lightweight, self-contained deployment in which all required models are loaded locally at runtime. This includes the merged fine-tuned TinyLlama model for response generation as well as the trained BERTopic model for topic assignment.

After the customer agent pastes a customer email, the system executes a fixed inference pipeline. First, the email text is processed by the BERTopic model to assign it to one of the predefined topic categories identified during topic modeling. In parallel, a short rule-based summary is generated by removing boilerplate expressions such as greetings and signatures and extracting the most informative sentences. Finally, the fine-tuned TinyLlama model generates a response draft conditioned on the original email, selected tone, and embedded compliance instructions. This architecture enables clear separation between topic analysis, summarization, and response generation while remaining computationally efficient.

B. User Interface and Interaction Flow

The user interface is designed to support a human-in-the-loop workflow rather than full automation. Customer service

agents paste incoming customer emails into a text input field and can choose between generating a formal or a friendly response. The interface displays both the assigned topic label and a short summary of the customer request alongside the generated reply. A confidence threshold is applied to topic assignments to reduce the risk of misleading classifications and to transparently communicate uncertainty.

Generated responses are automatically segmented into three components: greeting, main body, and closing. This structure allows agents to easily review, edit, or recombine individual parts of the draft before final use. A copy-to-clipboard function is provided to facilitate seamless transfer of the final response into existing customer service systems. Overall, the interface prioritizes usability and transparency, enabling agents to benefit from AI-assisted drafting without requiring direct interaction with the underlying code.

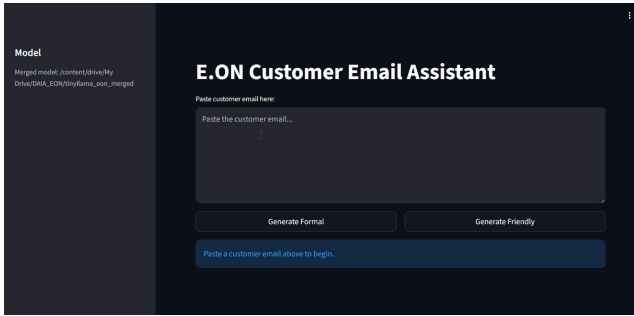


Fig. 7. Streamlit-based web application deployed within the Google Colab environment.

X. CONCLUSION & FUTURE OUTLOOK

A. Conclusion

This project explores how a compact, open-source large language model can be adapted to support customer service email drafting under realistic technical and organizational constraints. By combining topic modeling, synthetic instruction tuning, and parameter-efficient fine-tuning, we implement an end-to-end prototype that generates structured, tone-controlled draft responses augmented with safety checks and post-training validation mechanisms.

The evaluation results indicate that the system is capable of producing coherent and stylistically appropriate drafts that remain closely grounded in the customer-provided information. At the same time, the limited dataset size, reliance on synthetic responses, and controlled evaluation setting highlight that the system is best understood as an exploratory proof of concept rather than a fully mature solution.

Overall, the findings suggest that lightweight LLM-based assistants can provide meaningful decision support for customer service agents when deployed with explicit constraints and human oversight, particularly in regulated environments where full automation is neither feasible nor desirable.

B. Theoretical Implications

This work adds to existing research on the practical adaptation of large language models to domain-specific tasks under

constrained conditions. The findings suggest that parameter-efficient fine-tuning, when combined with explicit prompt-level guidance, can allow compact language models to handle structured drafting tasks without the need for large-scale proprietary systems. Moreover, the integration of topic modeling, rule-based preprocessing, and post-training validation highlights the value of hybrid NLP pipelines.

C. Business Implications

From a business perspective, the findings indicate that LLM-based drafting assistants can meaningfully support customer service operations without full automation. By accelerating response drafting while preserving human oversight, such systems have the potential to reduce handling time, improve consistency, and lower cognitive load for customer service agents. Importantly, the use of compact open-source models and lightweight validation mechanisms suggests that these benefits can be achieved under modest computational budgets, making the approach viable for enterprise deployment where cost, compliance, and transparency are key considerations.

D. Limitations and Future Work

While the proposed system demonstrates the practical feasibility of LLM-assisted customer service drafting, several limitations should be acknowledged.

First, the language model used in this project is intentionally compact. The fine-tuned TinyLlama model is selected to reflect realistic computational and budgetary constraints rather than to maximize absolute performance. While the results show that a small open-source model can generate coherent, tone-controlled, and factually grounded draft responses, larger and more recent foundation models would likely achieve higher linguistic fluency and robustness, particularly for complex or ambiguous customer inquiries.

Second, the instruction-tuning dataset is based on synthetically generated responses rather than authentic customer-agent reply pairs. This approach is necessary due to data availability, privacy, and compliance constraints. However, synthetic responses may not fully capture the stylistic diversity, operational shortcuts, and edge cases present in real-world agent communication, which may limit generalization.

Third, although the system is explicitly designed for human-in-the-loop usage, it has not yet been evaluated in a real operational setting with customer service agents. As a result, no empirical measurements of agent acceptance, editing effort, trust, or productivity gains are available. Such qualitative and quantitative user studies would be required to assess real-world usability.

Fourth, the evaluation was conducted on a relatively small validation set and focuses primarily on technical metrics such as training stability, lexical diversity, semantic similarity, and hallucination risk. While these measures provide important insights into model behavior and safety, they do not replace large-scale field evaluations.

Finally, the system does not incorporate retrieval-augmented generation (RAG) or access to internal enterprise systems. This

design choice prioritizes safety and compliance by preventing the generation of personalized or account-specific information, but it also limits the depth of automation that can be achieved without having access to actual factual and helpful information.

Future work could address these limitations by incorporating agent-edited responses into the training loop, evaluating larger language models under comparable constraints, integrating retrieval-based factual grounding, and conducting human-in-the-loop studies to assess productivity, quality, and trust in real customer service environments.

REFERENCES

- [1] Ada Support. (2024). Real-world examples of generative AI for customer service. Ada. <https://www.ada.cx/blog/real-world-examples-of-generative-ai-for-customer-service/>
- [2] Boston Consulting Group. (2023, July 6). How generative AI is already transforming customer service.
- [3] Bird, S., Klein, E., & Loper, E. (2009). *Natural language processing with Python*. O'Reilly Media.
- [4] Cai, W., Jiang, J., Wang, F., Tang, J., Kim, S., & Huang, J. (2025). A survey on mixture of experts in large language models. *IEEE Transactions on Knowledge and Data Engineering*, 37(7), 3896–3915. <https://doi.org/10.1109/TKDE.2025.3554028>
- [5] Costa, C., & Ghosh, S. (2025). Empowering customer service with generative AI: Enhancing agent performance while navigating challenges. *Information Research*, 30(iConf). <https://doi.org/10.47989/ir30iConf47566>
- [6] Dahl, S., & Gustafsson, R. (2025). *Evaluating BERT for email text classification: A comparative study with manual text pattern matching* (Bachelor's thesis, Stockholm University). <http://www.diva-portal.org/smash/get/diva2:1970734/FULLTEXT01.pdf>
- [7] DeepSeek-AI. (2024). DeepSeek-V3 technical report. *arXiv*. <https://doi.org/10.48550/arXiv.2412.19437>
- [8] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 4171–4186). Association for Computational Linguistics. <https://doi.org/10.18653/v1/N19-1423>
- [9] Dettmers, T., Pagnoni, A., Holtzman, A., & Zettlemoyer, L. (2023). QLoRA: Efficient finetuning of quantized LLMs. *arXiv*. <https://doi.org/10.48550/arXiv.2305.14314>
- [10] E.ON SE. *Financial Statements Pursuant to German GAAP and Combined Group Management Report for the 2024 Financial Year*. Essen, Germany: E.ON SE, 2024.
- [11] Farquhar, S., Kossen, J., Kuhn, L., & Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*, 630, 625–630. <https://doi.org/10.1038/s41586-024-07421-0>
- [12] Forbes. “Sales revenue of largest electric utility companies worldwide 2025 (in billion U.S. dollars).” in *Statista*, Jun. 12, 2025. [Online]. Available: <https://www.statista.com/statistics/274670/biggest-electric-utilities-in-the-world-based-on-sales/>. Accessed: Nov. 23, 2025.
- [13] Google. (2026). *Gemini 3 (Gemini 3 Flash version)* [Large language model]. <https://gemini.google.com/>
- [14] Grootendorst, M. R. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*. <https://doi.org/10.48550/arXiv.2203.05794>
- [15] Guo, Y., Shang, G., & Clavel, C. (2024). Benchmarking linguistic diversity of large language models. *arXiv*. <https://doi.org/10.48550/arXiv.2412.10271>
- [16] Holtzman, A., Buys, J., Du, L., Forbes, M., & Choi, Y. (2020). The curious case of neural text degeneration. *arXiv*. <https://arxiv.org/abs/1904.09751>
- [17] Hu, E. J., Shen, Y., Wallis, P., Li, Z., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-rank adaptation of large language models. *arXiv*. <https://doi.org/10.48550/arXiv.2106.09685>
- [18] K. M. S. Islam, R. T. Karri, S. Vegesna, J. Wu, and P. Madiraju, “Contextual embedding-based clustering to identify topics for healthcare service improvement,” in *Proc. 2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)*, 2025, pp. 794–799. Available: <https://arxiv.org/abs/2504.14068>
- [19] Janssens, W., Bogaert, M., & Van den Poel, D. (2025). LLM-assisted topic reduction for BERTopic on social media data (Version 1). *arXiv*. <https://doi.org/10.48550/arXiv.2509.19365>
- [20] Kannan, A., Kurach, K., Ravi, S., Kaufmann, T., Tomkins, A., Miklos, B., Corrado, G., Lukacs, L., Ganea, M., Young, P., & Ramavajjala, V. (2016). Smart Reply: Automated response suggestion for email (Version 1). *arXiv*. <https://doi.org/10.48550/arXiv.1606.04870>
- [21] Klarna Group plc. (n.d.). *About us*. <https://www.klarna.com/>
- [22] Li, S., Stenzel, L., Eickhoff, C., & Bahrainian, S. A. (2025). Enhancing retrieval-augmented generation: A study of best practices (Version 1). *arXiv*. <https://doi.org/10.48550/arXiv.2501.07391>
- [23] Liu, Y. (2024). Comparison of LDA and BERTopic in News Topic Modeling: A Case Study of The New York Times’ Reports on China. *Pacific International Journal*, 7(3), 47–51. Retrieved from <https://rclss.com/pij/article/view/616>
- [24] Mangrulkar, S., Gugger, S., Debut, L., Belkada, Y., Paul, S., & Bossan, B. (2022). PEFT: State-of-the-art parameter-efficient fine-tuning methods [Computer software]. GitHub. <https://github.com/huggingface/peft>
- [25] McInnes, L., Healy, J., & Astels, S. (2017). hdbSCAN: Hierarchical density-based clustering. *Journal of Open Source Software*, 2(11), 205. <https://doi.org/10.21105/joss.00205>
- [26] McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for dimension reduction. *arXiv preprint arXiv:1802.03426*. <https://doi.org/10.48550/arXiv.1802.03426>
- [27] Meister, C., Vieira, T., & Cotterell, R. (2020). Best-first beam search. *arXiv*. <https://arxiv.org/abs/2007.03909>
- [28] OpenAI. (2025, August 13). *GPT-5 system card*. <https://openai.com/index/gpt-5-system-card/>
- [29] OpenAI. (2025). *ChatGPT (GPT-5.2)* [Large language model]. <https://chat.openai.com/>
- [30] Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., & Lowe, R. (2022). Training language models to follow instructions with human feedback. *arXiv*. <https://doi.org/10.48550/arXiv.2203.02155>
- [31] Pan, J., Deng, S., & Huang, S. (2025). CoAT: Chain-of-associated-thoughts framework for enhancing large language models reasoning (Version 3). *arXiv*. <https://doi.org/10.48550/arXiv.2502.02390>
- [32] Pascanu, R., Mikolov, T., & Bengio, Y. (2013). On the difficulty of training recurrent neural networks. *arXiv*. <https://doi.org/10.48550/arXiv.1211.5063>
- [33] PyTorch Foundation. (n.d.). *Dropout*. PyTorch Documentation. <https://docs.pytorch.org/docs/stable/generated/torch.nn.Dropout.html>
- [34] Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing* (pp. 3982–3992). Association for Computational Linguistics. <https://aclanthology.org/D19-1410/>
- [35] SAP SE. (n.d.). *About SAP*. <https://www.sap.com/>
- [36] SentenceTransformers. (n.d.). *all-MiniLM-L6-v2*. Hugging Face. <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>
- [37] Siino, M., Tinnirello, I., & La Cascia, M. (2024). Is text preprocessing still worth the time? A comparative survey on the influence of popular preprocessing methods on Transformers and traditional classifiers. *Information Systems*, 121, 102342. <https://doi.org/10.1016/j.is.2023.102342>
- [38] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1), 1929–1958.
- [39] Streamlit. (2023). *Streamlit: The fastest way to build and share data apps* [Computer software]. <https://streamlit.io>
- [40] Thiergart, J., Huber, S., & Übellacker, T. (2021). Understanding emails and drafting responses: An approach using GPT-3 (Version 3). *arXiv*. <https://doi.org/10.48550/arXiv.2102.03062>
- [41] TinyLlama. (2024). *TinyLlama-1.1B-Chat-v1.0* [Large language model]. Hugging Face. <https://huggingface.co/TinyLlama/TinyLlama-1.1B-Chat-v1.0>

- [42] Ünlü, C. (2023). InterpreTutor: Using large language models for interpreter assessment. In *Proceedings of the International Conference on Human-Informed Translation and Interpreting Technology (HiT-IT 2023)*, Naples, Italy. https://doi.org/10.26615/issn.2683-0078.2023_007
- [43] Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., Davison, J., Shleifer, S., von Platen, P., Ma, C., Jernite, Y., Plu, J., Canwen, X., Le Scao, T., Gugger, S., & Rush, A. M. (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (pp. 38–45). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2020.emnlp-demos.6>
- [44] Wulf, J., & Meierhofer, J. (2024). Utilizing large language models for automating technical customer support (Version 1). *arXiv*. <https://doi.org/10.48550/arXiv.2406.01407>
- [45] Zhang, P., Zeng, G., Wang, T., & Lu, W. (2024). TinyLlama: An open-source small language model. *arXiv*. <https://doi.org/10.48550/arXiv.2401.02385>